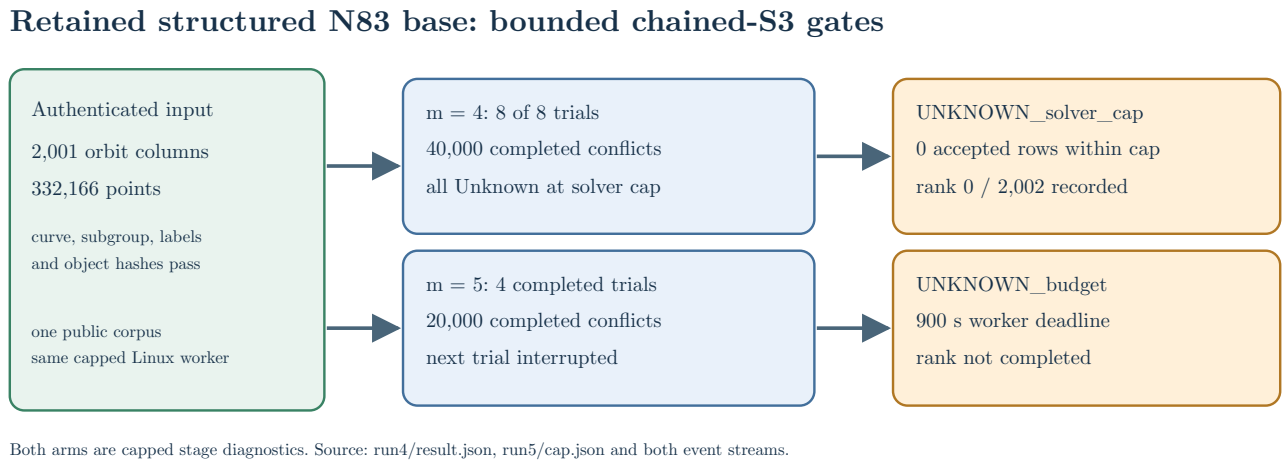
Retained structured N83 base: bounded m = 4 query and m = 5 control

Date: 2026-10-17 UTC. **Curve:** `icv1-f2m83-tm6151469093347-debefd74`. The frozen protocol is `PROTOCOL.md`; full receipts and commands are in `RUN_REPORT.md`. The signed-Frobenius archive passed the native import checks: 2,001 orbit columns, 332,166 points, 166,083 distinct x-coordinates.



Both arms are capped stage diagnostics. Source: run4/result.json, run5/cap.json and both event streams.

Bounded observations

Arm	Completed calls	Completed conflicts	Final receipt
m = 4	8 / 8	40,000	`UNKNOWN_solver_cap`
m = 5	4 / 8	20,000	`UNKNOWN_budget`

All completed calls returned `Unknown` at 5,000 conflicts. The m = 4 worker exited 0 in 784.331 s L0, with zero accepted rows within the cap, rank 0 of 2,002 columns, and 487,276,544 bytes cgroup peak. The m = 5 worker reached its 900-second own deadline during the next call and exited 124 without OOM. Its unfinished call has no completed conflict count; rank and cgroup peak were not written.

Provenance and interpretation

The same Linux arm64 executable served both arms, SHA-256 `a26a3aeda02c70cf732c9219f4eb09cb8be3a65afd8f03cad74f86bd0e9bc477`. Source commit: `1b00cf8a3450540aacd7778b6b2115276f46a81c`. Each container had one CPU, 4 GiB memory, zero swap and no network. The worker enforced 900 seconds, 8 trials, 5,000 conflicts per query and 32 models per query. The shared host was L0; a separate exporter build overlapped part of the m = 5 arm. The arms ask different decomposition questions, so their wall times are not a speed ratio.

The m = 4 exact results are `run4/result.json`, `run4/events.jsonl` and `run4.container-state.json`. The m = 5 receipts are `run5/cap.json`, `run5/events.jsonl` and `run5.container-state.json`. No accepted relation was available for independent relation replay. No useful-rank or full-cold cost comparison is established. The current-head library build obstruction is recorded in `problems/current-main-library-build-20261017.md`.

The next gate compares a retained K = 2,048 explicit-orbit base with this base under m = 4 and a degree-83 planted relation control. A larger K = 16,627 prebuilt base is the separate cardinality-driven construction candidate. Canonical progress graphs and the IC scoreboard remain unchanged.